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Using Structural Analysis to Inform Peer Support Arrangements for High School Students With Severe Disabilities

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Abstract

We examined the effects of peer support arrangements as an alternative to exclusive direct support from paraprofessionals for three high school students with severe disabilities in general education classrooms. We also explored the use of structural analysis as a data-based approach for further refining the interventions to enhance their impact in particular classrooms. Peer support arrangements were associated with substantial increases in social outcomes; academic engagement maintained or improved modestly for all participants. Moreover, structural analyses yielded findings used to adapt peer support arrangements to address students' individual needs. Although the effects of structural analysis-based adaptations resulted in increases in social responses for only one participant, levels of social interactions maintained, and improvements in academic engagement occurred for two participants.

Key Words: *peer-mediated; peer support arrangements; inclusion; adolescents; structural analysis*

Over the past 25 years, much empirical attention has focused on identifying effective pathways for supporting the inclusion of students with severe disabilities in general education classrooms (McDonnell & Hunt, 2014; Ryndak, Jackson, & White, 2013). Ensuring students with significant cognitive impairments can meaningfully access the social and learning opportunities within inclusive classrooms has been a particular challenge within secondary schools. A constellation of barriers can converge to limit meaningful participation for these adolescents (Carter, Bottema-Beutel, & Brock, 2014). For example, the complex communication, behavioral, and learning challenges experienced by students with severe disabilities can limit social opportunities. The attitudes and knowledge of peers can lead to their reluctance to spend time with classmates with severe disabilities (Carter, Common, et al., 2014). The heavy emphasis on lecture within many secondary classrooms can limit opportunities for students with and without disabilities to engage in conversation, shared work, or mutual assistance (Carter et al., 2016). In addition, the extensive reliance on paraprofessional support models can inadvertently create physical and social barriers to social interaction.

Peer support arrangements provide an evidence-based pathway for addressing these barriers within inclusive middle and high school classrooms. These peer-mediated interventions involve identifying one or more peers to provide ongoing academic, social, and behavioral support to a student with severe disabilities throughout the semester (Carter, Moss, Hoffman, Chung, & Sisco, 2011). "Peer partners" are recruited from within the same classroom, attend an initial orientation session regarding their roles, provide personalized supports outlined in a written plan, and receive needed guidance from paraprofessionals within the classroom. Each of these intervention components can be individually tailored to address the unique needs of a particular student and the classroom context in which he or she is enrolled.

Numerous studies have demonstrated the advantages of peer support arrangements relative

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to prevailing support models that rely extensively or exclusively on one-to-one paraprofessionals. This body of research includes at least eight single-case studies (Biggs, Carter, & Gustafson, 2017; Brock, Biggs, Carter, Cattey, & Raley, 2016; Brock & Carter, 2016; Carter et al., 2011; Carter, Sisco, Melekoglu, & Kurkowski, 2007; Cushing, Kennedy, Shukla, Davis, & Meyer, 1997; Shukla, Kennedy, & Cushing, 1998, 1999) and one randomized controlled trial (Carter et al., 2016). In terms of social outcomes, this collection of studies indicates higher levels of peer interaction—as well as increases in the initiations and responses of students—are evident when students with severe disabilities receive support primarily from peer partners rather than individually assigned adults. Likewise, academic engagement has either maintained or increased when students with severe disabilities worked alongside their peer partners. Such gains are accompanied by strong affirmation of the acceptability and feasibility of these interventions among students and staff.

Although available research clearly demonstrates that peer support interventions are more effective than "business-as-usual" support models, little attention has focused on possible pathways for individualizing these interventions for particular students. Students with severe disabilities are quite heterogeneous, and the general education classrooms they attend vary widely in terms of their instructional and social contexts. The next important stage in this line of research involves identifying data-based approaches for individualizing peer-mediated interventions in ways that maximize their impact for a particular student in a given classroom. Component and parametric analyses offer one approach for identifying how different intervention configurations influence student outcomes. For example, Carter, Cushing, Clark, and Kennedy (2005) observed higher levels of social interaction and curricular access when middle school students with severe disabilities worked with two peers relative to one peer. Similarly, Shukla et al. (1998) examined the impact of the adult supervision component of the intervention on the active engagement of peer partners. Both studies suggest different configurations of these interventions may indeed lead to differential outcomes for participating students. Although peer support arrangements can support active social and academic participation in general education classrooms, additional approaches are needed to examine how their impact can be strengthened further by attending to how various features are implemented.

Structural analysis holds particular promise as an avenue for carrying out this individualization. Structural analysis is the experimental analysis of contextual variables (i.e., antecedents, setting events, or other environmental variables) that may increase or decrease the likelihood a behavior will occur. Strong empirical support exists for the use of structural analysis as a means of assessment for decreasing problem behavior and increasing appropriate behavior. However, few studies have examined the use of structural analysis to develop interventions designed to increase social behaviors for students with severe disabilities. Peck, Sasso, and Jolivette (1997) conducted structural analyses in special education classrooms to test the effects of various contextual variables (e.g., peer interactions, task structure, noise level) on the social interactions and on-task behavior of five elementary school students. Likewise, Stichter, Randolph, Kay, and Gage (2009) involved peers in structural analyses to examine the effects of various contexts on prosocial (e.g., interactions and engagement) and problem behaviors (e.g., touching peers) of three elementary school participants with autism across general education settings. Boyd, Conroy, Mancil, Nakao, and Alter (2007) used structural analysis to understand the effects of circumscribed interests on social behaviors of young children with autism in preschool and kindergarten.

Unlike functional analysis, for which there is a suggested set of conditions to test, a preferred set of conditions does not exist for structural analysis. The possible antecedents, setting events, or environmental factors that may influence a particular behavior can be extensive, which may provide one reason for the lack of research on implementation of structural analysis in general education classrooms. Given the challenge of identifying all potentially relevant contextual factors related to specific behaviors within complex classroom environments, the application of this assessment procedure to just those contextual variables relevant to an intervention would narrow down the number of variables to consider. By focusing only on contextual variables relevant to peer support arrangements, hypotheses about which variables may increase or decrease social behaviors can be evaluated more easily.

The purpose of this study was to (a) evaluate the effectiveness of peer support arrangements in general education classrooms, and (b) explore the

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use of structural analysis as a means of refining this intervention for high school students with severe disabilities in those same classrooms. Our research questions include: Are peer support arrangements effective in increasing the social initiations and responses of students with severe disabilities relative to an exclusive reliance on paraprofessional support? To what extent is structural analysis a feasible and informative means of assessing the effects of contextual variables related to peer support arrangements within general education classrooms? How does the addition of structural analysis-based components to peer support arrangements impact social initiations and responses? How do paraprofessionals, teachers, peers, and students view the acceptability and impact of peer support arrangements and structural analysis procedures?

Method

Participants and Setting

After receiving university and district approval, we recruited participating students and staff from two suburban high schools in Tennessee through special education case managers.

Students with severe disabilities. We invited students who: (a) had a primary special education category of autism or intellectual disability; (b) were eligible for the state's alternate assessment for students with significant cognitive impairments; (c) were included in at least one general education classroom, other than physical education; and (d) had a paraprofessional or special educator present in the general education class to provide support. Six students met these criteria; we excluded one due to frequent absences, one who moved, and one who already had a peer support arrangement.

John was a 15-year-old White male with an intellectual disability and language impairment in ninth grade. He was enrolled in just one general education class, during which he received curriculum modifications (e.g., directions in multiple formats, modified difficulty and grading scale of assignments and tests, extra prompts on assignments, reduced distractions). According to school records, his full-scale IQ was 50 on the Stanford-Binet Intelligence Scale. John communicated verbally using full sentences, but his interactions prior to the study were primarily directed to other students with severe disabilities in the class. These interactions often were off-task and age-inappropriate (e.g., tickling, calling out to a peer during class). His 50-min World History class had 27 ninth-graders, and John was one of five students with disabilities. One paraprofessional provided supplemental support for all five students.

Tyler was a 15-year-old White male with autism in the ninth grade. According to school records, his full-scale IQ was 75 on the Universal Non-verbal Intelligence Test (UNIT), and he scored in the mild-moderate range on the Childhood Autism Rating Scale-Second Edition (Schopler, Van Bourgondien, Wellman, & Love, 2010). He was enrolled in four general education elective classes and had curriculum modifications for reading and math. Tyler's speech intelligibility made him difficult to understand. He would repeat himself when asked, speaking more slowly and clearly. He interacted primarily with other students with disabilities prior to the study and would frequently engage in other preferred activities (e.g., writing lists or letters, drawing pictures) rather than assigned tasks. The study took place in his Nutrition class, which lasted 54 min. We chose this class based on the consistent availability of a paraprofessional who could facilitate the intervention. The class had 19 students across grade levels, including five students with disabilities. Two paraprofessionals were assigned to support specific students.

Richard was a 15-year-old White male with autism. He was in the ninth grade and had no recent intelligence test information, but records indicated he was unable to complete tasks following standardized directions for the UNIT. Richard communicated using gestures and verbally, typically in two- to three-word utterances. He was enrolled in three general education elective classes (i.e., Art, Computer Science, Physical Education) and received various accommodations (e.g., frequent breaks, a predictable schedule, discrete directions in alternate formats, a choice of reinforcers). Prior to the study, Richard interacted almost exclusively with the two paraprofessionals in the classroom. He engaged in high rates of scripting behavior, mostly referencing movies, which was occasionally loud and disruptive. The study took place in his Art class, which lasted 50 min. We selected Richard's Art class for this study, rather than Computer Science, to better accommodate the scheduling of observations. The class included 29 students of varying grade levels, including three students with severe disabilities.

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Paraprofessionals. We recruited the paraprofessionals assigned to each student as intervention facilitators. John's paraprofessional had 15 years of experience and was responsible for providing support to John and four other students with disabilities in the class. She had implemented a peer support arrangement the prior year. Tyler's paraprofessional was in her first year and had no prior experience with students with disabilities. She supported one additional student with a disability in the class. Richard's paraprofessional provided one-to-one support. Although she had 3 years experience as an early childhood paraprofessional, it was her first year working in high schools.

Peer partners. Two or three peers without severe disabilities in each class were invited by paraprofessionals to be "peer partners" within peer support arrangements. Paraprofessionals consulted with the student with severe disabilities and general educator to identify peers the focus student already knew or had a preference for, who had consistent attendance, who were considered to be positive role models, and who were willing to help others. Every peer who was invited to participate agreed to do so. All peer partners reported being familiar with the focus students, but none were observed to have extended interactions during the baseline phase.

John's peer partners were both 14-year-old, White males in his same grade; both were "peer buddies" in middle school. Three 17-year-old seniors worked with Tyler. One White male and one Hispanic female began at the start of the first intervention phase; a third peer partner (a White female) joined during the second intervention phase based on the structural analysis findings. Two of the three peers had been peer partners earlier in high school. Three peers were recruited in Richard's Art class. A 17-year-old senior had previously served as a peer partner. The others–a 15-year-old Hispanic male and a 16-year-old White female–were both in 10th grade and had no prior experience working with students with severe disabilities.

Dependent Measures

We conducted direct observations two to three times per week using a paper-pencil recording system and a vibrating interval timer set to 15 s intervals. We used a single data sheet for all variables, which indicated the specific interval (i.e., 15 s, 30 s, 5 min) during which each measure was to be observed and recorded. Observations averaged 44 min in duration (range, 23-55 min depending on school schedule changes), started at the beginning of each class period, and lasted the entire class. A coding manual is available by request.

Social initiations were any verbal or nonverbal (e.g., gestures, signs) behavior directed to or from the student with severe disabilities and proceeded by at least 5 s without interactive behavior with the same student. Observers also coded initiations if the focus student interacted with a new peer. *Responses* were verbal or nonverbal communicative behaviors that directly followed (i.e., within 5 s) and corresponded to another student's initiations. Observers coded initiations and responses separately for the student with a disability, peer partners, and other classmates. We did not code interactions with other students with severe disabilities or school staff. Because peer partners were not yet identified during the baseline phase, all peer interaction partners were coded as other classmates. We coded initiations and responses using 15-s partial interval recording (i.e., 15-s observe, 15-s record). We collapsed initiations and responses into social interactions during the structural analyses only. This reflected our interest in maximizing any type of interaction with peers and prevented our having to examine all conditions separately for each measure.

We coded the *quality* of social interactions using a 3-point, Likert-type scale. Our quality measure was the average of scores for *content* (i.e., 3=appropriate, 2=neutral, 1=inappropriate), *reciprocity* (i.e., 3=high, 2=medium, 1=low), and *affect* (i.e., 3=positive, 2=neutral, 1=negative). Observers rated content as appropriate when focus students' interactions were age-appropriate and contextually consistent with the conversation. Inappropriate content was dramatically different from same-age peers and unrelated to the conversational context; interaction content was neutral when there was a mix of appropriate and inappropriate content. Reciprocity was high when interactions were reciprocal and balanced (both the focus student and peers equally initiated and responded), medium when either the focus student or peers initiated more frequently but responses were balanced, and low when the focus student rarely initiated and infrequently responded to peers' initiations. Observers coded affect according to facial expressions and body language, coding positive affect based on smiles and attentive body language (e.g., leaning in); negative affect based on frowns, expressions of displeasure, or physical or

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verbal aggression; and neutral affect if interactions were neither positive or negative. At the end of each 5-min interval, observers provided a summary rating of the content, reciprocity, and affect based on all interactions occurring in the 5-min interval, unless no interactions took place.

We defined *academic engagement* as the focus student looking at materials (e.g., textbook, worksheet, overheads) related to ongoing instructional activities; looking at the teacher; writing related to the assigned activity; following teacher directions; raising hand; or asking questions of the teacher, paraprofessional, or another student about instructional activities. We operationally defined *proximity to peers* as having a body orientation, distance (i.e., no more than 5 ft), and position by which the focus student could readily interact with at least one peer without disabilities. The definition of *proximity to paraprofessional* was being within 5 ft of the focus student. We used momentary time sampling to measure *academic engagement* and *proximity*, recorded every 30 s.

Peer comparisons. To estimate typical levels of social initiations and responses and academic engagement, we used the same observational measurement system to collect normative data on a sample of peers from each participant's classroom. We completed three peer comparison observations per classroom during both the baseline and first peer support intervention phases. Observers randomly selected peers who did not have severe disabilities and coded one peer for the first half of the class period and another for the second half. We established the normative range of initiations, responses, and academic engagement as 0.5 *SD* above and below the mean.

Interobserver Agreement

Three masters-level research assistants and the first author conducted all observations. Observers reached a minimum of 90% reliability across three videos and three live practice sessions prior to conducting observations. We evenly distributed reliability sessions throughout the study and across phases. We calculated interobserver agreement (IOA) by dividing the number of intervals with agreement by the total number of intervals (i.e., agreements plus disagreements) and multiplying by 100%. We collected IOA during 37.6% of classroom observations and averaged 97.7% agreement (range, 85.8–100%) across all measures. Table 1 includes IOA for each observational measure. We collected IOA during 35.6% of structural analysis sessions and averaged 99.2% for social interactions (range, 86.0–100%).

Social Validity

We assessed social validity at the conclusion of the study. Paraprofessionals, general educators, and peer partners completed a survey comprised of Likert-type questions. Each questionnaire included 20–23 items (see Table 3 for a complete list of items). Students with disabilities completed a different 8-item questionnaire with the option to respond *yes*, *no*, or *I don't know*, as well as provide open-ended responses. John and Tyler completed the measure independently; the paraprofessional read the questions to Richard and documented his responses.

Experimental Design and Procedures

We used a multiple probe across participants design to examine the effectiveness of the peer support arrangements, as well as the addition of the structural analysis-based components for increasing social initiations and responses. (The

Table 1
Interobserver Agreement (IOA) for All Observational Measures

Measure	*M* (range)
Focus student interactions	
Initiations	95.5% (92.1–100%)
Responses	93.2% (85.8–100%)
Peer partner interactions	
Initiations	95.4% (91.7–98.6%)
Responses	93.2% (85.9–99.2%)
Other peer interactions	
Initiations	96.4% (89.3–100%)
Responses	95.9% (92.7–100%)
Quality	
Content	99.6% (88.0–100%)
Reciprocity	94.0% (25.0–100%)
Affect	98.7% (75.0–100%)
Academic engagement	97.5% (96.6–98.1%)
Proximity to paraprofessional	97.9% (89.0–100%)
Proximity to peer partners	99.1% (98.8–99.7%)
Proximity to other peers	99.0% (98.8–99.7%)

Note. Cells display percentage of interobserver agreement and ranges for each observational measure.

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experimental design for the structural analysis is described later.) We used visual analysis of initiations and responses to make phase change decisions. Once a clear change was demonstrated in the first intervention phase, we embedded structural analyses into each peer support arrangement to determine the effects of specific contextual variables on the occurrence of social interactions. Before introducing the second intervention phase (i.e., adapted peer supports), we collected additional data to confirm no changes in social interactions occurred as a result of the structural analysis procedure, the passage of time, or some other unknown variable.

Baseline phase. Each student with disabilities received direct support from his assigned paraprofessional within the classroom. Although the general educator was responsible for delivering instruction to all students in the class, the paraprofessional assisted with implementing modifications or accommodations and supporting the focus student's participation in class activities. All three focus students sat in close proximity to peers without disabilities in the class. The assignments given to each student were aligned with the content other students in the class were learning. During this phase, paraprofessionals did not receive any additional guidance or training, but continued supporting students with severe disabilities as they had previously. Table 2 addresses the extent to which intervention components took place during the baseline phase.

Peer support intervention. Procedures were consistent with Carter et al. (2016).

Paraprofessional training and peer support planning. Prior to recruiting peers, each parapro-

Table 2
Summary of Treatment Fidelity Across Participants and Study Phases

	John		
Abbreviated fidelity indicators	Base line	Peer support	Peer support +
Mean number of peer partners present	—	2.0	2.3
Peer partners are in proximity to focus student	**0%**	**100%**	**100%**
Sat next to each other	100%	100%	100%
Remained in proximity during out-of-seat activities	n/a	100%	n/a
Joined the same group during group activities	n/a	100%	100%
Peer partners interacted with the focus student	0%	100%	100%
Greeted the focus student	0%	100%	100%
Engaged in conversation	0%	100%	100%
Included the student in interactions with other peers	0%	100%	100%
Peer partners assisted focus student academically	**0%**	**100%**	**100%**
Helped the student participate in class activities	0%	100%	100%
Repeated or rephrased instructions for the student	0%	100%	67%
Appropriately prompted the student	0%	100%	100%
Provided appropriate feedback to the student	0%	100%	100%
Worked together on classroom activities	0%	100%	100%
Shared work materials	0%	100%	100%
Facilitator supported peer partner and student	**0%**	**100%**	**100%**
Facilitated interactions during class when appropriate	0%	100%	100%
Provided reminder/feedback to peer partners before, during, or after class	0%	100%	67%
Provided praise and feedback to students during or outside of class	0%	100%	100%
SA-based components implemented as planned	—	—	**100%**
SA-based component 1	—	—	100%
SA-based component 2	—	—	100%

Note. SA = structural analysis. Peer Support + denotes the adapted peer supports phase.

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fessional participated in an initial training lasting 1.5 to 2 hr ($M = 1.8$ hr). The first author provided individual didactic training for each paraprofessional, consisting of oral instruction, guided discussion, and collaborative development of an individualized peer support plan. Each training closely followed printed manuals (available by request) addressing the goals of the intervention, strategies for recruiting peers, creating peer support plans, orienting and supporting peers, fading support, and the role of the intervention coach. Each paraprofessional received a binder, which included information from the training and supplemental materials (e.g., scripts for recruiting peers, sample peer support plans). Immediately after training, paraprofessionals collaborated with the coach on a written peer support plan specific to the student and classroom. Each plan addressed individualized goals for the student, facilitation strategies for the paraprofessional (e.g., highlighting similarities among the students, providing positive feedback for working together, redirecting interactions to peer partners), and support strategies for the peer partners (e.g., ensuring the focus student has a role in group activities, encouraging interactions with other classmates, highlighting important information in notes or on a worksheet). Each plan included strategies appropriate to each instructional format (e.g., whole group, small group, individual seatwork) used in the class. Although she received training on the peer support intervention prior to this study, John's paraprofessional also participated in the same individual didactic training and peer support planning provided to all paraprofessionals as part of this study.

Table 2
Extended

Tyler			Richard		
Base line	Peer support	Peer support +	Base line	Peer support	Peer support +
—	2.0	2.7	—	3.0	2.5
50%	**100%**	**100%**	**0%**	**100%**	**100%**
0%	100%	100%	0%	100%	100%
n/a	100%	n/a	n/a	n/a	n/a
n/a	100%	100%	n/a	n/a	n/a
50%	100%	100%	0%	75%	100%
50%	100%	100%	0%	100%	100%
0%	100%	100%	0%	100%	100%
0%	50%	0%	0%	0%	0%
0%	**100%**	**100%**	**0%**	**50%**	**100%**
0%	100%	100%	0%	25%	50%
0%	50%	67%	0%	0%	0%
0%	50%	67%	0%	25%	0%
0%	100%	100%	0%	50%	100%
0%	100%	100%	0%	0%	0%
0%	100%	100%	0%	25%	0%
0%	**100%**	**100%**	**0%**	**100%**	**100%**
0%	50%	67%	0%	100%	100%
0%	100%	100%	0%	75%	100%
0%	50%	33%	0%	25%	0%
—	—	**100%**	—	—	**100%**
—	—	100%	—	—	100%
—	—	100%	—	—	100%

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Table 3
Paraprofessional and Teacher Perspectives on Acceptability of Interventions and Assessment

	John		Tyler		Richard	
Questionnaire items	PA	GE	PA	GE	PA	GE
Overall, I enjoyed being in this project.	5	4	5	4	5	5
I feel I was effective in this role.	5	—	4	—	4	—
The student with a disability benefitted *socially* from having a peer support.	5	5	4	5	5	5
The student with a disability benefitted *academically* from having a peer support.	5	4	4	4	3	4
The peers without disabilities benefitted *socially* from being a peer support.	5	5	4	5	5	5
The peers without disabilities benefitted *academically* from being a peer support.	4	4	3	3	3	3
I am motivated to continue using this strategy.	5	3	4	4	5	5
The amount of time required to use this strategy was reasonable.	5	5	4	5	3	5
I would need ongoing consultation to keep implementing this strategy.	1	2	2	2	2	3
Implementation of this strategy required considerable support from other school staff.	2	3	4	1	4	3
I would not be interested in implementing this strategy again. *	1	3	2	1	1	1
The assessment results were useful to further understand the needs of the student.	4	4	4	4	4	4
This strategy fits well within this classroom.	5	4	4	5	5	5
I understood the procedures of this strategy.	5	4	4	4	5	5
I would know what to do if I was asked to implement this strategy again.	5	3	4	4	4	4
Participation in the assessment required a considerable amount of time.*	2	2	3	1	2	2
The student with a disability has more friends as a result of this project.	5	5	4	4	5	4
The assessment process would be helpful for other students that I work with.	5	4	4	4	5	4
This strategy was a good way to address the educational needs of the student with a disability.	5	3	4	5	5	5
This strategy negatively impacted other students in the class. *	2	2	2	1	1	1
I could use the strategies I learned through this project with other students.	5	4	3	4	5	4
I often use cooperative learning strategies with students in my classroom.	—	4	—	5	—	3
The peer support strategy would be feasible for me to implement if additional school staff were not in my classroom.	—	3	—	5	—	4
This strategy was a good way to address the educational needs of students without disabilities.	—	4	—	3	—	4
The amount of time required for record keeping with this strategy was reasonable.	5	—	4	—	5	—

Note. PA = Paraprofessional. GE = General educator. 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, 5 = strongly agree. *Negatively worded items on which lower scores indicate endorsement.

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Peer partner training. Paraprofessionals delivered the initial orientation for all peer partners, which lasted 15 to 25 min ($M = 21.7$ min). These meetings were held during class at times designated by the general educator to avoid missing instruction. Each student with a disability was asked if he wanted to participate; none did. Paraprofessionals reviewed topics according to a written outline: (a) rationale for peer support strategies; (b) background about the focus student (e.g., personal interests, communication needs); (c) general goals of increasing the number of peers with whom the student with severe disabilities interacts, increasing involvement in classroom activities, and decreasing reliance on adult support; (d) confidentiality and respectful language; (e) review and discussion of the written peer support plan; and (f) guidance about when to seek assistance. Peers were encouraged to share ideas or ask questions.

Facilitation of peer support arrangements. Following peer training, the students sat in proximity to one another. Paraprofessional facilitation of each peer support arrangement included modeling interactions, prompting students to greet each other and interact throughout the class, identifying opportunities for social interactions, encouraging the students to work together on assignments when appropriate, and reinforcing peer partners and students with disabilities for working together and interacting. Paraprofessionals self-monitored their use of facilitation strategies and documented academic and social support strategies used by peer partners (e.g., shared activities, use of appropriate prompting and feedback, engaging in conversation) and proximity of the student with a disability and peer partners at least weekly using fidelity checklists provided in the intervention manual.

Coaching. The first author served as a coach in each intervention phase. Coaching occurred one to two times in the first week and continued weekly throughout the intervention. Each coaching session lasted an entire class period. Initially, coaching involved modeling of facilitative behavior and in-the-moment feedback/prompting for the paraprofessional. After the first week, the coach faded her support to feedback provided only before and/or after the class session; later, she checked in and problem solved only when needed. The coach provided support for facilitative behaviors such as pointing out opportunities for interactions to peer partners, redirecting focus student interactions from adults to peer partners, identifying shared interests, and distinguishing opportunities for academic support appropriate for peer partners. We kept the fading of coach support consistent across all tiers and planned to provide increased support only if measures of fidelity dropped below 80%. Richard's paraprofessional met the criteria for additional coaching. She increased and maintained her fidelity after one regularly scheduled coaching session, and we provided no additional coaching sessions.

Structural analysis procedures. After an initial demonstration of effect for peer support arrangements for each participant, the first author conducted a structural analysis in collaboration with paraprofessionals and the involvement of peers.

Selection of contextual variables. First, the first author and paraprofessional collaborated to identify a list of contextual variables hypothesized to contribute to higher or lower levels of initiations and/or responses or account for some of the variability in the initial peer support data. Paraprofessionals drew on their firsthand observations of the peer support arrangement and their background knowledge of the students to identify times when they seemed to interact well (or when interactions seemed difficult). We asked paraprofessionals to use this information to list all variables they thought might contribute to students' ease of interaction and those thought to impede it. We excluded variables not feasible to control without substantial interference in the daily routine of the class, such as instructional format (e.g., whole-group instruction vs. small-group activities) and physical arrangement of the classroom (e.g., location of desks and work spaces). We next selected two contextual variables we anticipated would most affect the level of social interaction in each peer support arrangement and developed hypotheses about each.

For John, we chose (a) the location of the peer support arrangement within the class and (b) proximity of the paraprofessional. We compared two locations of the peer support arrangement—sitting at desks in rows amidst other students or sitting at a small table toward the rear of the room. We hypothesized social interactions would be higher when students were seated at the small table because it allowed students to gain each others' attention and interact without turning around in their seats or leaning across the aisle. For proximity of the paraprofessional, we observed social behavior when she was in close proximity (i.e., within 5 ft) and when she was away. In both conditions, the

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paraprofessional was available if John requested academic support. If John requested assistance from the paraprofessional during the low-proximity condition, she would approach him briefly and respond according to the training (see next section). We hypothesized close proximity of the paraprofessional would inhibit social interactions with peers, as John often directed academic questions to adults before peers when they were available. To control for variability of interactions caused by the presence of other untrained peers, we seated two additional peer confederates in proximity to the peer support arrangement in both conditions. Peer confederates were told to respond in a neutral manner if the focus student spoke with them, but not to initiate conversations with him during sessions.

The contextual variables for Tyler were (a) number of peers in his work group and (b) type of task assigned to him. Specifically, we examined levels of social interaction when Tyler was working with one peer partner verses two or three peers (i.e., either two peer partners or one or two peer partners and a peer confederate). We hypothesized Tyler would engage in high levels of interaction when paired with one peer. For type of task, we compared mastered tasks (i.e., tasks he could work on for a least 3 min independently) and unmastered tasks (i.e., tasks for which he required support at least every 1 min). Mastered tasks included cutting magazine pictures, copying words or numbers directly from a list, and drawing pictures. Unmastered tasks included searching for information on the Internet, locating information in text or notes, and following a recipe. We hypothesized Tyler would engage in higher levels of interactions with unmastered tasks, as we thought he would check in with peers more frequently to gain assistance.

For Richard's structural analysis, the contextual variables were (a) choice of peer partner and (b) choice of task. For the first, he was given the choice of the peer partner with whom he wanted to work or he was assigned a peer partner. To control for peer choice during the assigned peer conditions, the peer partner assigned in a peer-choice condition was not assigned in the next no-choice condition. Because Richard appeared to prefer the two female peers, we hypothesized the peer choice condition would result in higher rates of interaction. As part of the structural analysis, we also compared assigned tasks (i.e., the same assignment as other students in the class with appropriate modifications) and tasks chosen by Richard (e.g., coloring sheets of plain white paper with markers or colored pencils, white board with markers). Richard typically required substantial support for assigned art projects and could complete chosen tasks independently. Therefore, we hypothesized he would interact at higher rates during the task choice condition as the paraprofessional could more easily fade support.

Alternating treatments design. To experimentally evaluate the effects of hypothesized contextual variables on levels of social interactions, we used an alternating treatments design for each structural analysis. Structural analysis sessions were 5 min in duration, with a minimum of 2 min between each session, and were conducted in the students' general education classrooms during ongoing class activities. We conducted sessions over a period of 4 or 5 days for each participant (i.e., sessions 23 to 28 for John, 36 to 39 for Tyler, and 39 to 42 for Richard). To minimize disruption, all condition changes were coordinated with the general educator. We collected data during the structural analyses using the same data collection system as all other phases for all participants, except Richard. Due to low levels of interactions during the first intervention phase and the possibility that partial interval recording could overestimate the actual occurrence of the behavior in conditions with low rates of interactions, we also collected data on the frequency of initiations and responses during the 15-s record interval to ensure data collection was sensitive enough to show changes in level across conditions. Results of Richard's structural analysis include frequency of interactions, rather than percent of intervals.

We trained paraprofessionals and peer partners on the structural analysis procedures. These trainings took 10-15 min and focused on (a) instructing paraprofessionals and peers to respond consistently to all social interactions across conditions (i.e., to keep the consequences of interactions constant) and (b) directing peers to follow cues to initiate interactions (i.e., to control for the number of opportunities for students with disabilities to respond). We directed each paraprofessional to respond to initiations directed to her in a brief, neutral way (i.e., neutral tone and facial expression and less than five-word responses, if possible). If the student requested help with an assignment, the paraprofessional would aid him as quickly as possible without drawing out the interaction. For example, if a student requested

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help finding the answer to a question on a worksheet, she would use gestures and minimal language to identify the key words in the question and direct him to the appropriate place in the text. Peer partners were provided the same guidelines for responding to the focus student's initiations. However, in Richard's case, peers were taught to respond with low-quality praise statements (e.g., "Thanks for telling me." "That sounds good.") and positive facial expressions (e.g., smiles) consistently across all conditions. Finally, peers were taught to follow cues provided by researchers to initiate interactions with the focus student. A limited list of brief, open-ended initiation statements (e.g., "How's it going?") were provided to peers to use during each session. During structural analysis sessions, the researcher stood at a distance, out of the line of sight of the focus student. She raised a hand to cue the peer to initiate an interaction with the focus student at 1-min intervals to ensure a minimum of five opportunities for the focus student to respond.

Adapted peer supports. To evaluate the effects of intervention modifications based on the results of each student's structural analysis, each peer support arrangement incorporated adaptations corresponding with the structural analysis conditions resulting in the highest levels of social interactions (i.e., structural analysis-based components; see Results section). All other aspects of the intervention remained the same in this second intervention phase. In John's History class, students sat at a small table toward the back of the classroom where they were still able to participate in instruction and classroom activities along with the rest of the class. The paraprofessional was instructed to continue to monitor, check in, and provide reinforcement and feedback, while minimizing proximity throughout the class. For Tyler, an additional peer partner was trained to ensure more than one peer would be available to work with him at a time. The paraprofessional monitored the work Tyler was assigned to ensure an appropriate level of difficulty, rather than already mastered tasks. Richard was given a textual prompt (i.e., a card that read "Will you work with me?") to choose the peer partner with whom he wanted to work. His paraprofessional prompted him to use it at least twice during each class period. Although still required to complete assigned tasks during his Art class, Richard could choose tasks during downtime (i.e., when waiting to start or when finished with his work) and request breaks.

Treatment Fidelity

Observers collected data on treatment fidelity across all participants and phases, including baseline (23.3%), peer support (35.1%), and adapted peer supports (29.6%). Fidelity data addressed paraprofessional facilitative behaviors (e.g., facilitating interactions, providing reinforcement and feedback), peer support behaviors (e.g., helping to participate in class activities, sharing materials, engaging in conversations), and maintaining close proximity of the students with disabilities and peer partners throughout the class. During the second intervention phase, fidelity measures included implementation of the structural analysis-based components. Overall fidelity, calculated as an average of the core components of the intervention (i.e., bolded items in Table 2), was 100% in the first and second intervention phase for John, 100% in the first and second intervention phase for Tyler, and 81.5% in the first and 100% in the second intervention phase for Richard.

Results

Figure 1 displays the effect of peer support arrangements and the addition of structural analysis components on our primary outcomes of social initiations and responses. Gray bars indicate the levels of peer comparison initiations and responses. Figure 2 displays academic engagement for students with severe disabilities, with gray bars indicating levels of academic engagement for peer comparisons. Academic engagement was a secondary outcome and was not considered as part of phase change or structural analyses decisions.

Peer Support Intervention

Relative to the peer comparison data collected in each classroom, we observed low, stable levels of focus student social initiations and responses, across all three tiers during baseline phases. The introduction of the peer support arrangements coincided with immediate increases in social initiations and responses, more closely approximating normative rates, for all three participants. Academic engagement improved or maintained at high levels for all three participants during the peer support intervention phase. Changes in level and increasing trends in social initiations and responses across all tiers indicated a functional

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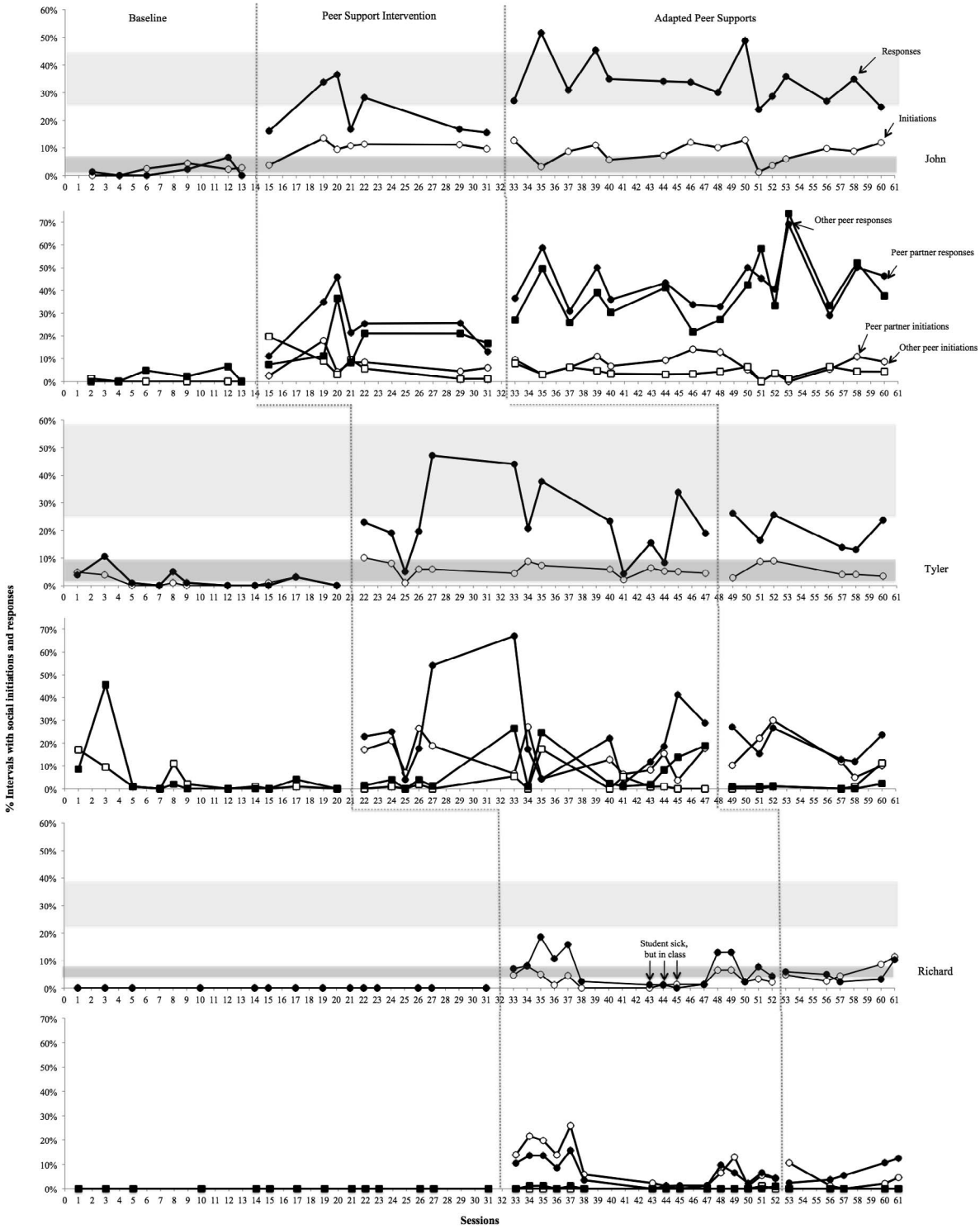


Figure 1. Initiations and responses of students with disabilities, peer partners, and other peers. Dark gray and light gray bars denote normative rates (i.e., 0.5 *SD* above and below the mean) of social initiations and responses, respectively, for each setting.

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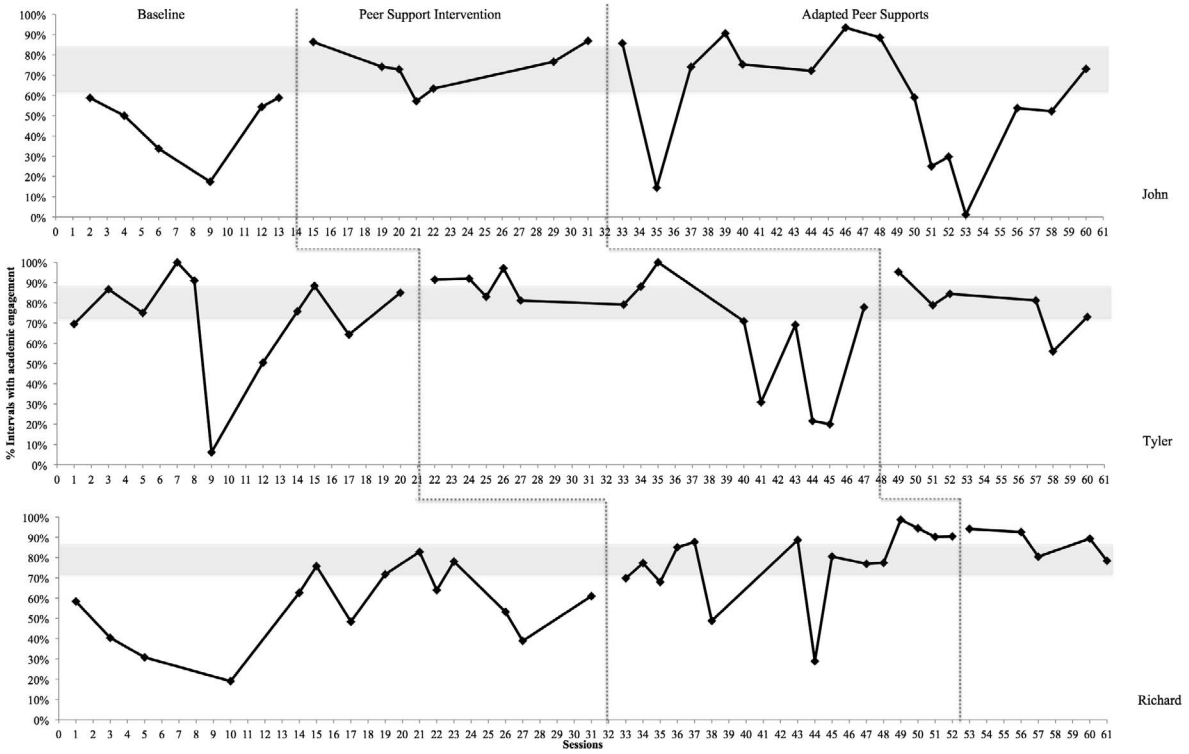


Figure 2. Academic engagement of students with disabilities. Gray bars denote normative rates (i.e., 0.5 *SD* above and below the mean) of academic engagement for each setting.

relationship between peer support arrangements and social initiations and responses.

John. Baseline levels of John's social initiations and responses averaged 1.9% and 1.7% of intervals, respectively. Other peers directed initiations and responses toward John in an average of 0.2% and 2.3% of intervals, respectively. Despite receiving prior training on the peer support intervention, John's paraprofessional did not implement any intervention components during the baseline phase, with the exception of ensuring proximity to peers. An immediate change in level of social initiations and responses occurred upon introducing peer partners, and an increasing trend was observed. Mean percent of intervals was 9.9% for initiations and 22.3% for responses, with 14.3% and 0% overlap (i.e., the percentage of data points in the intervention phase that fell within the range of all data points in the baseline phase), respectively. Initiations were slightly higher and responses slightly lower than normative data. With the introduction of the peer support arrangement, average quality ratings increased from 2.1 to 2.9 for appropriateness, 1.3 to 2.5 for reciprocity, and 1.8 to 2.6 for affect. Although an increasing trend occurred across two data points immediately preceding the phase change, we saw an immediate increase in level of academic engagement and an increase in means from 45.5% of intervals to 73.9% in the first intervention phase, at or above engagement of peer comparisons.

Tyler. Tyler's social initiations and responses remained below 10% across all data points in the baseline phase except one. After introducing peer supports, we observed an immediate and overall increase in level for both initiations ($M = 5.8\%$) and responses ($M = 23.1\%$). Although social initiations and responses were highly variable (i.e., 23.1% overlap for initiations and responses), the level of Tyler's initiations compared to those of his classmates, and average responses were slightly lower. Improvement in the quality of interactions was limited to reciprocity, which increased from a mean rating of 1.5 to 2.2. Ratings of content maintained at high levels ($M = 3.0$), and average ratings of affect decreased from 2.7 to 2.6. High rates of academic engagement, similar to classmates, maintained following the initial implementation of the peer support intervention.

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We observed increased variability over time. However, a week-long school break occurred between session 41 and 43, which may have decreased academic engagement.

Richard. Although in proximity to peers during an average of 93.4% of intervals, Richard had no initiations and responses across baseline observations. Introduction of the peer support arrangements resulted in increases in both social initiations ($M = 3.2\%$) and responses ($M = 7.0\%$), and quality ratings were 2.2 for content, 1.3 for reciprocity, and 2.6 for affect. Whereas the increases in social initiations and responses were relatively substantial, low occurrence of both behaviors was observed during four probes (across sessions 43-47) following a week-long break from school. Although the data indicated the intervention was implemented with an appropriate level of fidelity, the paraprofessional reported that Richard appeared to be sick for at least three of the four days. Despite low levels of interactions on these days, overlap of data points was 13.3% for initiations and 6.7% for responses. Richard's academic engagement increased from 56.1% to 77.5%, more closely approximating his classmates.

Structural Analysis

Figure 3 shows results of structural analyses for each participant.

John. The results of John's structural analysis indicated higher levels of social interaction occurred during both low-proximity conditions (i.e., whole group and small group). This confirmed our hypothesis regarding increased social interactions given low proximity of the paraprofessional, but not our hypothesis that social interactions would be higher when students sat in a small group at the back of the room. Although the differentiation between these conditions was inconsistent and small, the structural analysis (SA)-based components incorporated into the peer support intervention were small group and low proximity of the paraprofessional.

Tyler. Differentiation among conditions in Tyler's structural analysis indicated higher levels of social interaction occurred during the small group plus unmastered task condition (see Figure 3). Our hypothesis that working one-on-one with a peer would result in higher levels of interactions was not confirmed. However, our hypothesis that we would observe high levels of interactions when Tyler worked on unmastered tasks was confirmed. Based on these results, the paraprofessional invited an additional peer partner so students could work in a small group. In addition, the paraprofessional monitored Tyler's work to ensure it was at an appropriate level of difficulty.

Richard. For Richard, the structural analysis indicated higher rates of social interaction occurred during the task-choice and peer-choice condition, confirming our original hypotheses. During the peer support intervention phase, Richard frequently requested particular activities and materials (e.g., paper and markers). Therefore, SA-based components included choice of activity or materials whenever possible. However, during early structural analysis conditions, Richard consistently struggled to communicate the peer with whom he wanted to work. Throughout the structural analysis, when asked whom he wanted to work with, the paraprofessional prompted him to point, saying, "Show me," to clarify his choice. During the subsequent intervention phase, we introduced a more socially appropriate textual prompt (i.e., a card reading, "Will you work with me, please?"). The paraprofessional prompted Richard to use the card at least twice during the class period.

Adapted Peer Supports

The addition of the SA-based components resulted in an increase in level of social responses for one participant, but although we observed decreased variability, levels of social initiations and responses did not improve for the other two participants. However, we observed improvements in academic engagement for these two participants. Although functional relations between adapted peer support arrangements and increased social interactions and academic engagement were not found, improvements in at least one of these outcomes occurred for each participant.

John. We observed an immediate change in level for John's responses ($M = 34.0\%$) but not initiations ($M = 8.3\%$). While the variability of responses decreased to a standard deviation of 8.5% and variability of initiations increased to a standard deviation of 3.7%, changes in variability were not substantial. Overlap of data points was 100% for initiations and 80% for responses. Mean quality measures decreased slightly for content and reciprocity after the addition of the SA-based components, but average measure of affect maintained. The variability of academic engagement ($SD = 29.3$) increased considerably during the second intervention phase and mean occur-

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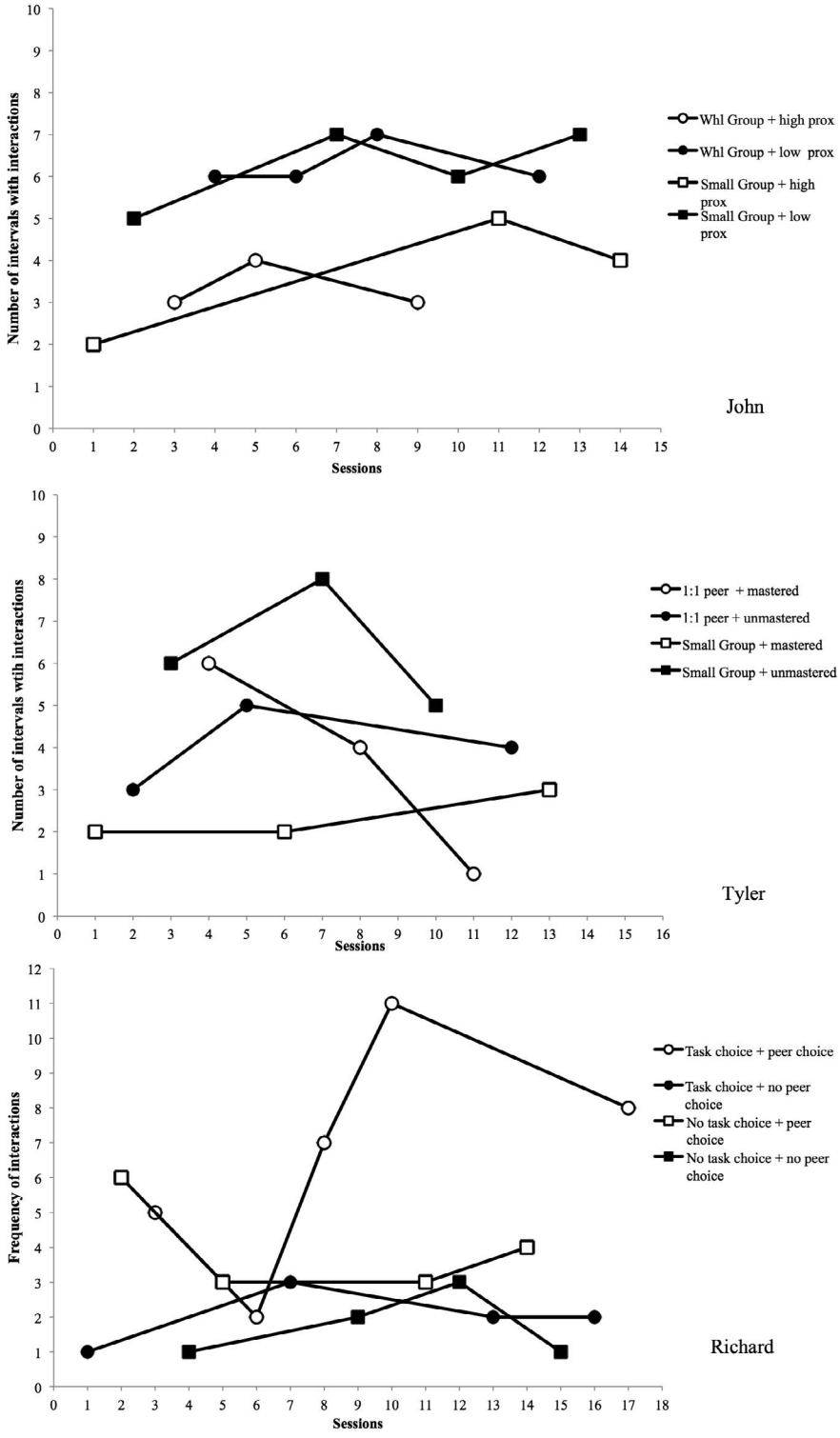


Figure 3. Structural analysis results for each participant. Top two panels (John and Tyler) display number of intervals with interaction. Bottom panel (Richard) displays frequency of interactions.

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rence decreased to 59.2% of intervals. However, the average percent of intervals with no instruction also increased from 19.6% in the first intervention phase to 28.3% with an increasing trend in the second phase, which may partially account for the decrease in academic engagement. We also observed these changes as adult proximity remained low ($M = 11.5\%$).

Tyler. While social initiations remained at comparable levels, variability of the responses decreased substantially from a standard deviation of 13.9% to 5.5%, suggesting the contextual variables identified in the structural analysis may have accounted for much of the variability. Quality of interactions maintained throughout this phase in terms of content ($M = 3.0$) and affect ($M = 2.6$), and ratings of reciprocity increased slightly ($M = 2.5$). In addition, average academic engagement increased from 71.1% to 78.1% with a decrease in variability ($SD = 12.0\%$).

Richard. The introduction of the SA-based components resulted in an increasing trend and overall change in level for Richard's initiations ($M = 6.3\%$), with 60% overlap of data points in the first intervention phase. These levels were consistent with peer comparisons. However, we observed a decrease in responses, averaging 5.3%. This corresponded with a decrease in peer partner initiations, from 9.0% to 3.7%, which meant fewer opportunities for Richard. Furthermore, the average ratings of the quality of his interactions increased on all dimensions, including content ($M = 2.7$), reciprocity ($M = 1.8$), and affect ($M = 3.0$). Mean academic engagement increased to 87.0%, while variability of engagement decreased ($SD = 7.1\%$). We also saw small increases in proximity to peer partners ($M = 91.2\%$) and other peers ($M = 97.1\%$). Proximity to paraprofessional increased as well ($M = 87.0\%$). Although Richard requested peers to work with him, we did not observe an increase in partners instructional format, which remained at 0% of intervals.

Social Validity

Both paraprofessionals and general educators reported the time required to implement the intervention was reasonable and strategies were appropriate for the classrooms and a good way to address the needs of the participants with disabilities (see Table 3). Paraprofessionals also indicated they knew what to do to implement the strategy again and felt they could use it with other students with disabilities. Paraprofessionals and general educators indicated the assessment procedure required a reasonable amount of time and the results were useful. Peers noted they felt effective in their role, would recommend being a peer partner to other students, and would do it again in the future (see Table 4). All peers considered the student with a disability to be a friend. All three participants said they enjoyed working with their peer partners, would like to continue to do so, learned new things, and considered their peer partners to be friends. In addition, all three participants' initiations, responses, and academic engagement more closely approximated peer comparisons with peer support arrangements relative to baseline.

Discussion

We examined the effectiveness of peer support arrangements for supporting inclusion at the high school level, as well as the role of structural analysis as a means of further individualizing this intervention. Our findings confirm and extend the literature in several ways.

First, we replicated the findings of prior research by demonstrating social improvements associated with peer support arrangements at the high school level (Carter et al., 2007, 2011, 2016). In addition, academic engagement improved or maintained at levels comparable to classmates for all participants. Prior to intervention, students with severe disabilities interacted with peers at extremely low rates, despite regular close proximity to numerous potential conversational partners. By shifting some support behaviors from the paraprofessional to peers, students with disabilities were better able to participate both academically and socially as part of the class. Normative data indicated these improved social interactions approximated those of peers in terms of both level and quality. These also extend the peer support literature by comparing social interactions of students with disabilities with those of their classmates.

Furthermore, peer support interventions resulted in improvements in academic engagement for one participant whose baseline level was below his classmates and maintenance of the already high levels of academic engagement for the other two participants. Paraprofessionals or general educators may be concerned the increase in social interactions or decrease in direct adult support may

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Table 4
Peer Partner Perspectives on Social Validity

	John		Tyler			Richard	
Questionnaire items	P1	P2	P1	P2	P3	P1	P2
At first, I was excited to become a peer support.	3	4	5	5	3	5	5
I felt confident serving in this role.	3	4	4	5	2	5	5
I had enough help from a teacher or teaching assistant (i.e., paraprofessional) to do this role well.	5	4	5	5	5	5	4
This was too much work for me.*	2	2	1	1	2	1	1
I feel I was effective in this role.	4	4	4	4	3	5	5
It was easy to get my own work done while part of this project.	4	4	4	5	4	5	3
The initial orientation meeting with a teacher/paraprofessional was helpful.	4	4	5	5	5	5	5
Other students in the class should also do this.	3	4	4	4	5	4	4
I would be a peer support again in the future.	3	4	5	4	4	5	3
I understand why the teachers thought peer supports would be helpful for my partner with a disability.	4	4	5	4	5	4	5
Our school should have more peer supports for students with disabilities.	4	4	5	5	4	4	5
My partner benefited *socially* from having a peer support (e.g., talks more with peers, has more friends).	4	4	5	4	4	5	4
My partner with disabilities benefited *academically* from having a peer support (e.g., participates more in class, learns new skills).	4	4	4	4	3	5	4
I benefitted *socially* from being a peer support.	3	4	5	4	4	5	3
I benefitted *academically* from being a peer support.	3	4	4	3	3	4	3
I consider my partner with disabilities to be a friend.	5	4	5	4	4	5	5
I would recommend being a peer support to my other friends.	4	4	5	5	4	5	5
My views about students with disabilities have changed for the better.	4	4	5	5	4	5	5
I also spend time with other students who have similar disabilities at my school.	3	4	5	3	3	5	5
Overall, I enjoyed being in this project.	4	4	5	5	4	5	4

Note. P1-P3 = Peer partners. 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, 5 = strongly agree. *Negatively worded items on which lower scores indicate endorsement.

negatively impact students' academic engagement. However, our findings indicate this is not necessarily the case and are further supported by general educator and paraprofessional feedback indicating peer support arrangements fit well in their classrooms and were not disruptive to other students. Given that the academic support needs of students with severe disabilities are so commonly used to justify the need for direct adult support, these findings and those of previous studies are particularly relevant for the viability of peer support models as alternatives to direct adult support in general education classrooms.

Second, we addressed the role of structural analysis as a means of further individualizing and refining peer support arrangements. This procedure enabled comparison of the influence of contextual variables on the social interactions of students with severe disabilities. Although the structural analyses were researcher implemented, this assessment approach was feasible to implement with the collaboration of peers and paraprofessionals in general education classrooms without substantial disruption to ongoing instruction and class activities. These assessments provided results used to inform and refine peer

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support arrangements. Few studies have used structural analysis in general education classrooms, and none have used structural analysis as a means of informing an intact intervention. It is important to note that our original hypotheses about which variables would result in higher levels of social interactions were not always accurate. This further reinforces the importance of data-based approaches to inform adaptations rather than anecdotal assumptions about factors which may (or may not) influence outcomes.

Third, we demonstrated that different configurations of peer support arrangements can have differential effects on student outcomes, suggesting that the use of structural analysis holds promise as a means of tailoring these interventions. With the addition of the SA-based components to peer support arrangements, we saw clear increases in social interactions for one participant and modest decreases in variability for the other two. Although we hypothesized that adaptations based on the contextual variables with higher rates during structural analyses would result in increased social interactions when integrated into peer support arrangements, we observed this with only one participant. The other two participants appeared to have somewhat more consistent, yet comparable, levels of social interaction across both intervention conditions. We suspect these divergent findings may be due to (a) our limiting the structural analysis to just two variables, although other contextual variables not considered also impacted focus students' social interactions; or (b) the impact of contextual variables we suspected of affecting social interactions but ruled out because they were not feasible to control in the classroom environment.

In this study, structural analyses focused on social interactions, not academic engagement. However, the effects of the adapted peer support intervention on academic engagement differed for the participants who showed no improvements in social outcomes and the participant whose social interactions increased. Likely, these outcomes are related to the adaptations targeting academic tasks. Presenting Tyler with slightly more challenging tasks may have increased his interest or required sustained attention or effort to complete. Similarly, Richard's motivation to remain engaged in tasks may have been enhanced by offering him a choice of materials or tasks. It also appeared Richard took longer to complete activities when working independently, rather than with frequent or intensive adult support, resulting in less downtime (i.e., no instruction). On the other hand, neither of John's adaptations addressed the tasks or activities assigned to him, and his academic engagement became more variable as social interactions increased. It is possible efforts to find contexts that maximize social interactions only might not enhance (and could possibly negatively impact) academic engagement. Although not entirely aligned with our expectations, these results demonstrated that, although peer supports prescribes an individualized plan for each student, these interventions can be further refined, but the potential impact of adaptations on all relevant outcomes is a necessary consideration.

Fourth, our ability to tailor interventions to specific students and instructional settings as part of the initial peer support plan, as well as changes made during the adapted peer supports phase, indicates the flexibility with which peer support arrangements can be implemented. Although we outlined the intervention components (i.e., peer recruitment and orientation, proximity, shared activities, ongoing monitoring and facilitation), the manner and degree to which these elements were executed differed slightly for each participant, as evidenced by variance in implementation fidelity data. For example, implementation of Richard's peer support arrangement relied more heavily on paraprofessional facilitation of social interactions during class than it did for Tyler, whose intervention was successful with more focus on paraprofessional feedback for peer partners. The adaptability of peer support arrangements adds to the social validity of the intervention, which is further enhanced by the addition of an assessment method that can inform appropriate, data-based adaptations.

Recommendations for Practice

Findings from this study have implications for educators responsible for supporting the inclusion of students with severe disabilities in high school general education classrooms. Our findings indicate that, without encouragement and purposefully planned adult support, students with severe disabilities rarely or inconsistently engage in social interactions with peers without disabilities, even when seated in close proximity to one another. However, given a relatively short (i.e., 1.5–2 hr) training and coaching, paraprofessionals provided the ongoing facilitation necessary to promote social interactions using peer support arrangements.

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Shifting paraprofessionals' role to one designed to support students as they work with classmates promotes a more inclusive experience for adolescents with severe disabilities and provides more opportunities to develop social connections.

Our findings also highlight a promising avenue for assessing the influence of contextual variables on the effectiveness of peer support arrangements in inclusive contexts. Although this study did not examine the effectiveness and feasibility of the implementation of structural analyses by school-based personnel, we outline procedures and considerations for effective implementation, which may provide guidance to those who wish to use structural analysis as a means of formative assessment for this and other interventions. Practitioners can develop hypotheses by reflecting on peer support arrangements to determine the contextual factors consistently present or absent during successful and unsuccessful social interactions. They can describe situations when peer support arrangements seemed to be going well and times when students seemed to be struggling to interact. Some factors to consider may be interaction partner(s), classroom or seating arrangements, type of activity or task, or materials selected. Also, if a single contextual variable is hypothesized to have significant impact over all others, the steps described in this study may also be followed to test one variable at a time, for a simpler and possibly shorter assessment phase. The potential of structural analysis for use in natural settings, with low-rate behaviors, and with prosocial behavior lends itself for strong consideration as a formative assessment option to individually tailor peer-mediated interventions targeting social outcomes, such as peer support arrangements.

Limitations and Future Research

Several limitations warrant future research. First, evaluation of the peer support intervention—as well as our structural analyses—focused primarily on social rather than academic outcomes. Although we included a measure of academic engagement, future research would benefit from the addition of a stronger measure of learning or access to academic content. Second, to enhance the feasibility of structural analyses in natural settings, we made some tradeoffs that could have impacted the precision of the structural analyses. Constraints associated with conducting the structural analyses in general education classrooms led to assessment of fewer contextual variables, shorter duration of each session, and fewer data points collected for each condition. Additional research is needed to determine the best balance of precision and ecological validity. Third, structural analyses only focused on the social behavior of students with disabilities. However, social interactions are also influenced by the behavior of communicative partners, which we did not take into account when identifying or assessing relevant contextual variables. In Richard's case, for example, task choice may have led to decreased opportunities for peers to interact, as Richard frequently engaged in ritualistic and repetitive behavior when engaged in preferred tasks. These behaviors may have been difficult for peers to interpret, redirect, or counter. Consideration and assessment of the effects of contextual variables on the social behavior of peers should be considered as part of future research. Fourth, the first author implemented structural analyses for all participants. The question remains: Which school staff member would be responsible for conducting or overseeing this type of assessment? Given the limited research in this area, little is known about the implementation of structural analysis by anyone other than trained researchers or clinicians. However, Stichter et al. (2009) demonstrated general and special educators can participate in and conduct structural analysis sessions with detailed instructions and visual cues from researchers to ensure fidelity. Research exploring the implementation of functional analyses by classroom teachers and special educators (e.g., Bloom, Lambert, Dayton, & Samaha, 2013; Kodak, Fisher, Paden, & Dickes, 2013) offers further support for these individuals to do so effectively with appropriate training.

Conclusion

Findings from this study add to the growing body of literature supporting the use of peers as an alternative to exclusive reliance on adult support in general education classroom. It is apparent—from baseline data in this study and others—that simply enrolling students with severe disabilities in general education classes does not equate to meaningful inclusion. Although many peer-mediated interventions include elements that can be individualized, the role of assessment is often overlooked. Amidst advocacy for evidence-based practices, the field must not lose focus on data-based individualization of instruction and sup-

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ports. Establishing evidence-based assessment procedures for use in informing the selection of outcomes and choosing and refining interventions is necessary to maintain a bridge between high-quality research-based practices that address the individual needs of students with disabilities. Although structural analysis shows promise, future research is needed to enhance procedures and demonstrate its effectiveness for informing this and other interventions.

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